B-coloring of grid graphs

Zhengxu Jiang[1] and Jiaao Li[1]

[1]School of Mathematical Sciences and LPMC, Nankai University, Tianjin 300071, China

Abstract

A B-coloring is a proper edge-coloring in which every 4-cycle is rainbow, and $q_B(G)$ is the minimum number of colors in such a coloring of G. We determine q_B for all torus grids and all cylindrical grids. A torus grid requires four colors precisely when both side lengths are even and the grid is not $C_4 \square C_{4k+2}$ for an integer $k \geq 1$; every other torus grid requires five colors. A cylindrical grid with even circumference requires four colors, while one with circumference $2n+1$, where $n \geq 1$ is an integer, requires four colors for path length at most n and five colors thereafter. We also complete the exceptional family of higher-dimensional even discrete tori, showing that the B-coloring number equals twice the dimension whenever the dimension is at least three. Finally, for a positive integer ℓ, we study ℓ-cylindrical grids, products in which exactly ℓ coordinate directions are cyclic. When the cycle factors are even, we determine q_B except for three explicit classes in which it is either the maximum degree or one more than the maximum degree.

Keywords. B-coloring; grid graph; cylindrical grid; torus grid; rainbow 4-cycle.

1 Introduction

All graphs in this paper are finite and simple. An edge-coloring is *proper* if incident edges receive distinct colors, and a set of edges is *rainbow* if its edges have pairwise distinct colors. A *B-coloring* of a graph G is a proper edge-coloring in which every 4-cycle is rainbow. The *B-coloring number* $q_B(G)$ is the minimum number of colors in a B-coloring of G. Let $\Delta(G)$ denote the maximum degree of G. Since the edges at a vertex of maximum degree must receive distinct colors, $q_B(G) \geq \Delta(G)$.

For a graph G, let $V(G)$ and $E(G)$ denote its vertex and edge sets. The Cartesian product of two graphs G and H, denoted by $G \square H$, is the graph with vertex set $V(G) \times V(H)$ in which two vertices (u, x) and (v, y) are adjacent if and only if either $uv \in E(G)$ and $x = y$, or $u = v$ and $xy \in E(H)$. Let P_n and C_n denote the path and cycle on n vertices, respectively. The products $P_m \square P_n$, $P_m \square C_n$, and $C_m \square C_n$ are called *rectangular grids*, *cylindrical grids*, and *torus grids*, respectively.

Early work on edge-colorings with rainbow 4-cycles treated hypercubes by Faudree, Gyárfás, Lesniak, and Schelp [6] and graph products by Faudree, Gyárfás, and Schelp [7]. Gyárfás and Sárközy [9] later required the edge-coloring to be proper, introduced the term B-coloring and the parameter $q_B(G)$, and related this parameter to the Brown–Erdős–Sós

$(7,4)$-conjecture [2]. Their work also discusses connections with the Ramsey-type problem of Erdős and Gyárfás [5] and with star edge-coloring [4].

Recent work has determined or bounded $q_B(G)$ for several graph classes. For planar graphs, the best current bounds are $q_B(G) \le 2\Delta(G)+6$ in general, $q_B(G) \le 2\Delta(G)+4$ when $\Delta(G) \ge 12$, and $q_B(G) \le 2\Delta(G)$ when $\Delta(G) \ge 38$; for outerplanar graphs, $q_B(G) = \Delta(G)$ when $\Delta(G) \ge 7$ [8, 12]. Further results concern maximal planar graphs of diameter two, subcubic graphs, and perfect proper edge-colorings of regular bipartite graphs with rainbow 4-cycles [3, 14, 11].

For grid graphs, the starting point is the work of Gyárfás and Sárközy [10] on Cartesian products of paths and cycles. Their results determine q_B for all Cartesian products of paths, including higher-dimensional rectangular grids. For a graph G and a positive integer d, write G^d for the Cartesian product of d copies of G, and let $Q_d = P_2^d$ denote the d-dimensional hypercube.

Theorem 1.1 (Gyárfás and Sárközy [10]). *Let G be a Cartesian product of paths. Then $q_B(G) = \Delta(G)$ unless G is isomorphic to $P_2 \square P_n$ for some integer $n \ge 2$, Q_3, or Q_5. In the exceptional cases,*

$$q_B(P_2 \square P_n) = 4, \qquad q_B(Q_3) = 4, \qquad q_B(Q_5) = 6.$$

Passing from a rectangular grid to a cylindrical grid and then to a torus grid makes one and then both coordinate directions periodic. Consequently, the parity of the cycle lengths plays a central role in the exceptional cases below.

Thus the rectangular-grid case is complete. Their results for products involving cycles leave more pronounced parity gaps. More generally, a Cartesian product of cycles is a *discrete torus*; it is *even* if every factor is an even cycle. For an integer $d \ge 1$, an even discrete torus with factor lengths $2n_1 \le \cdots \le 2n_d$, where the $n_i \ge 2$ are integers, is *exceptional* [10] if $n_1 = 2$ and $2n_i \equiv 2 \pmod 4$ for every $i \in \{2,\dots,d\}$. Equivalently, an exceptional even discrete torus has the form $C_4 \square C_{2n_2} \square \cdots \square C_{2n_d}$, where $n_2,\dots,n_d$ are odd integers.

Theorem 1.2 (Gyárfás and Sárközy [10]). *Let $d \ge 1$ be an integer, and let $2n_1 \le \cdots \le 2n_d$ be the factor lengths of an even discrete torus, where the $n_i \ge 2$ are integers. If this discrete torus is non-exceptional, then*

$$q_B(C_{2n_1} \square \cdots \square C_{2n_d}) = 2d.$$

If it is exceptional, then

$$2d \le q_B(C_{2n_1} \square \cdots \square C_{2n_d}) \le 2d+1.$$

Moreover, $q_B(C_4) = 4$ and $q_B(C_4 \square C_{4k+2}) = 5$ for every integer $k \ge 1$.

For products of odd cycles, Gyárfás and Sárközy obtained the following bounds.

Theorem 1.3 (Gyárfás and Sárközy [10]). *Let $d \ge 1$ and $n_1,\dots,n_d \ge 1$ be integers, and set $G = C_{2n_1+1} \square \cdots \square C_{2n_d+1}$. Then $2d+1 \le q_B(G) \le 3d$. If every factor length is divisible by $2d+1$, then $q_B(G) = 2d+1$. Furthermore,*

$$2d+1 \le q_B(C_3^d) \le \begin{cases} 5d/2, & d \text{ is even}, \\ (5d+1)/2, & d \text{ is odd}. \end{cases}$$

Theorem 1.3 leaves a substantial gap when its divisibility condition fails. In particular, the preceding results do not determine a general torus grid with an odd side length. Our first main result closes this gap and, together with Theorem 1.2, gives a complete classification of torus grids.

Theorem 1.4 (Torus grids)**.** *For all integers $m, n \geq 3$,*

$$q_B(C_m \square C_n) = \begin{cases} 5, & \text{at least one of } m, n \text{ is odd}, \\ 5, & \{m, n\} = \{4, 4k+2\} \text{ for some integer } k \geq 1, \\ 4, & \text{otherwise.} \end{cases}$$

The new part of Theorem 1.4 is that one odd side length always forces exactly five colors, with no divisibility hypothesis. Our second main result gives the corresponding complete classification for cylindrical grids. Even circumference always permits four colors. For odd circumference, the answer changes exactly when the path becomes longer than half of the cycle.

Theorem 1.5 (Cylindrical grids)**.** *Let $m \geq 2$ be an integer. For every integer $n \geq 2$,*

$$q_B(C_{2n} \square P_m) = 4.$$

For every integer $n \geq 1$,

$$q_B(C_{2n+1} \square P_m) = \begin{cases} 4, & 2 \leq m \leq n, \\ 5, & m \geq n+1. \end{cases}$$

We next turn to higher dimensions. The analysis of torus grids also completes the exceptional family of even discrete tori. The cases $d = 1$ and $d = 2$ appear in Theorem 1.2; the new assertion is $q_B = 2d$ in every dimension $d \geq 3$.

Theorem 1.6 (Exceptional even discrete tori)**.** *Let $d \geq 1$ be an integer, and let $2n_1 \leq \cdots \leq 2n_d$ be the factor lengths of an exceptional even discrete torus, where the $n_i \geq 2$ are integers. Then*

$$q_B(C_{2n_1} \square \cdots \square C_{2n_d}) = \begin{cases} 4, & d = 1, \\ 5, & d = 2, \\ 2d, & d \geq 3. \end{cases}$$

We finally consider products of paths and cycles with at least one path factor. For a positive integer ℓ, a Cartesian product of paths and cycles with exactly ℓ cycle factors is an *ℓ-cylindrical grid* [1]; equivalently, it has ℓ periodic directions. A cylindrical grid is the two-dimensional 1-cylindrical case, and an ℓ-cylindrical grid is *even* if all of its cycle factors are even. To avoid duplicate representations, we replace every pair $P_2 \square P_2$ by a C_4 factor and represent the path and cycle parts as

$$P = P_2^r \square P_{m_1} \square \cdots \square P_{m_s}, \qquad E = C_{2n_1} \square \cdots \square C_{2n_t},$$

where $r \in \{0, 1\}$, $s \geq 0$, and $t \geq 1$ are integers, the m_i are integers satisfying $m_i \geq 3$ for $i = 1, 2, \ldots, s$, and the n_i are integers satisfying $n_i \geq 2$ for $i = 1, 2, \ldots, t$, with $2n_1 \leq \cdots \leq 2n_t$. Thus $G = P \square E$ is a t-cylindrical grid, and $r + s \geq 1$ ensures that it has a path factor. Let $c_4(E)$ denote the number of C_4 factors of E. The next theorem is uniform across dimensions: its two-dimensional cases agree with Theorem 1.5, while its new content concerns higher-dimensional grids.

Theorem 1.7 (Even ℓ-cylindrical grids). *Let $G = P\square E$ be as above, and assume $r + s \geq 1$.*

(1) If $P \cong P_2$ and $t = 1$, then $q_B(G) = 4$.

(2) If $P \cong P_2$ and $E \cong C_4\square C_4$, then $q_B(G) = 6$.

(3) In each of the following cases, $\Delta(G) \leq q_B(G) \leq \Delta(G) + 1$:

(a) $P \cong P_2$, $c_4(E) = 0$, and $t \geq 2$;

(b) $P \cong P_2$ and $E \cong C_4\square C_4\square C_{4n+2}$ for some integer $n \geq 1$;

(c) $P \cong P_2\square P_m$ for some integer $m \geq 3$ and $c_4(E) = 0$.

(4) In all remaining cases, $q_B(G) = \Delta(G)$.

Section 2 records the product tools used throughout. Sections 3 and 4 determine the B-coloring numbers of torus and cylindrical grids, respectively. We then turn to higher dimensions: Section 5 treats exceptional even discrete tori, and Section 6 proves the even ℓ-cylindrical-grid classification. The final section isolates the two open problems left by the higher-dimensional results.

2 Preliminaries

We first fix notation for the product constructions used throughout. For integers $d \geq 1$ and $n_1, \ldots, n_d \geq 2$, the product $P_{n_1}\square\cdots\square P_{n_d}$ is a d-dimensional grid. Given graphs G and H and a vertex $x \in V(H)$, let $G(x)$ denote the copy of G in $G\square H$ indexed by x, and define copies of H analogously. When a factor is a cycle, we read its coordinate modulo the length of that cycle.

The cross inequality below combines B-colorings of two factors, but its hypotheses also involve their chromatic numbers. These are easy to control for the products used here by Sabidussi's theorem; as usual, $\chi(G)$ denotes the chromatic number of G.

Theorem 2.1 (Sabidussi [13]). *For all graphs G and H,*

$$\chi(G\square H) = \max\{\chi(G), \chi(H)\}.$$

The edge-coloring component is the following proper form of the product construction in [6]; see also [10].

Lemma 2.2 (Cross inequality). *Suppose that G has a p-color B-coloring and that H has an r-color B-coloring. If $\chi(G) \leq r$ and $\chi(H) \leq p$, then*

$$q_B(G\square H) \leq p + r.$$

Proof. Use disjoint color sets A and B, with $|A| = p$ and $|B| = r$, and fix B-colorings α of G and β of H. Identify each palette with a cyclic group, and fix proper vertex colorings of G and H using $\chi(G)$ and $\chi(H)$ colors, respectively. For the vertex colors of H, choose distinct cyclic shifts of A; this is possible because $\chi(H) \leq p$. For the vertex colors of G, choose distinct cyclic shifts of B; this is possible because $\chi(G) \leq r$.

Color the copy $G(x)$ by the shift indexed by the vertex color of $x \in V(H)$, and color copies of H analogously. The coloring is proper because the two factor directions use disjoint palettes and each copy receives a permutation of a proper edge-coloring. A 4-cycle contained in a single factor copy is rainbow. Every other 4-cycle is a product square. Its two G-edges are copies of the same edge of G indexed by adjacent vertices of H, and hence receive distinct colors; the same holds for its two H-edges. Since A and B are disjoint, the square is rainbow. □

The following improvement for Q_3 will be used repeatedly.

Lemma 2.3 (The Q_3 product lemma [6, 10]). *If H is a connected bipartite graph with at least two edges, then*

$$q_B(Q_3\square H) \le q_B(H) + 3.$$

Consequently, if $q_B(H) = \Delta(H)$, then $q_B(Q_3\square H) = \Delta(Q_3\square H)$.

Proof. The inequality is the special recoloring construction for Q_3 from [6]; its proper version is stated in [10]. If $q_B(H) = \Delta(H)$, the upper bound is $\Delta(H) + 3$, which equals $\Delta(Q_3\square H)$ and hence is also a lower bound. □

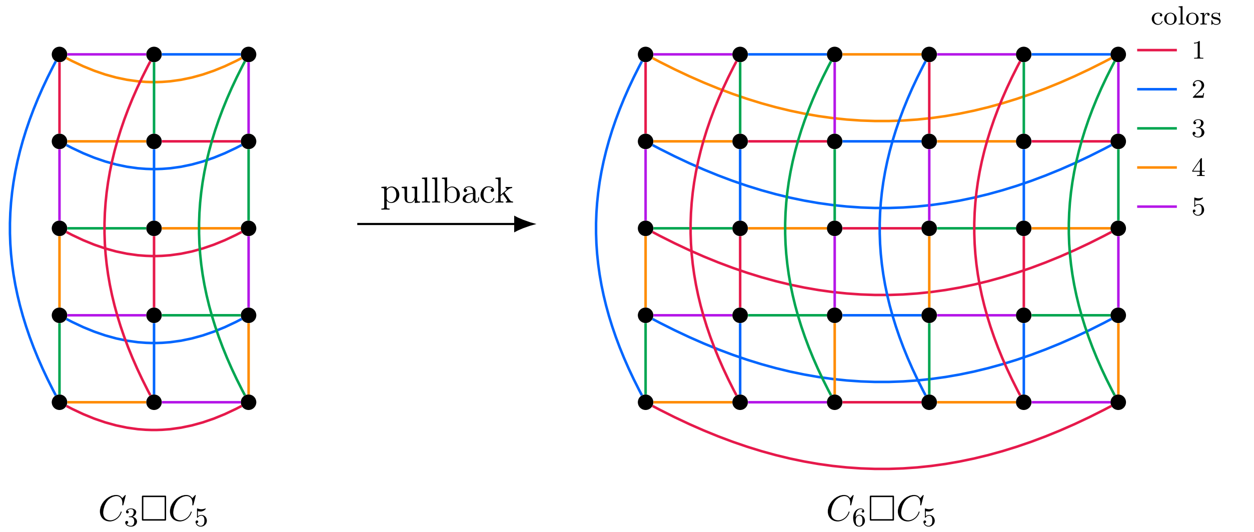

Figure 1. A five-color B-coloring of $C_3\square C_5$ and its pullback to $C_6\square C_5$. The legend at upper right identifies the five edge colors.

A graph map $\pi : G' \to G$ is a covering if, for every $x \in V(G')$, the edges incident with x are mapped bijectively to the edges incident with $\pi(x)$.

Lemma 2.4 (Covering pullback). *Let $\pi : G' \to G$ be a graph covering that maps every 4-cycle of G' onto a 4-cycle of G. If G has a k-color B-coloring, then so does G'.*

Proof. Let c be a B-coloring of G and define $c'(e') = c(\pi(e'))$. The covering property maps the edges incident with each vertex $x' \in V(G')$ bijectively to the edges incident with $\pi(x')$, so c' is proper. If Q' is a 4-cycle of G', then $\pi(Q')$ is a 4-cycle of G and is rainbow under c. Hence Q' is rainbow under c'. □

For every integer $n \ge 1$, the map $C_{4n+2} \to C_{2n+1}$ given in cyclic coordinates by $i \mapsto i \pmod{2n+1}$ is a double covering. Figure 1 illustrates the pullback from $C_3\square C_5$ to $C_6\square C_5$; corresponding lifted edges have the same color.

3 Torus grids with an odd factor

Theorem 3.1 (Torus grids with an odd side). *For all integers $m \geq 3$ and $n \geq 1$,*

$$q_B(C_m \Box C_{2n+1}) = 5.$$

The proof of Theorem 3.1 begins with the lower bound.

Lemma 3.2. *For all integers $m \geq 3$ and $n \geq 1$, the torus grid $C_m \Box C_{2n+1}$ has no four-color B-coloring.*

Proof. Suppose that such a coloring exists. Use coordinates $\mathbb{Z}_m \times \mathbb{Z}_{2n+1}$, and denote by $h(i,j)$ and $v(i,j)$ the colors of $(i,j)(i+1,j)$ and $(i,j)(i,j+1)$, respectively. Fix a color c, and for $i \in \mathbb{Z}_m$ let

$$A_i = |\{j : h(i,j) = c\}|, \qquad B_i = |\{j : v(i,j) = c\}|.$$

At every vertex, the four incident edges have the four colors. Summing over column i gives

$$A_{i-1} + A_i + 2B_i = 2n + 1. \tag{3.1}$$

Similarly, every square contains color c exactly once, and summing over the squares between columns i and $i+1$ gives

$$2A_i + B_i + B_{i+1} = 2n + 1. \tag{3.2}$$

Comparing (3.1) and (3.2), with indices shifted when necessary, yields $A_i + B_i = A_{i-1} + B_{i-1}$. Thus $A_i + B_i = T$ is independent of i. Substituting $B_i = T - A_i$ and $B_{i+1} = T - A_{i+1}$ into (3.2) gives $A_{i+1} - A_i = 2T - (2n+1)$. Since the indices are cyclic, this constant difference is zero. Hence $2T = 2n + 1$, a contradiction. □

To attain the lower bound, we construct five-colorings. We begin with $C_4 \Box C_{2n+1}$.

Table 1. Blocks for the five-coloring of $C_4 \Box C_{2n+1}$. In each column, Y_i lists the colors on the C_4 copy and X_i lists the four edges to the next column.

	B_3		B_4		B_5	
column	Y_i	X_i	Y_i	X_i	Y_i	X_i
0	1234	3542	1234	5451	1234	5451
1	2315	4153	2143	1532	2143	4532
2	5421	2315	3214	5143	3251	5413
3			4521	2315	1342	4251
4					5423	2315

Lemma 3.3. *For every integer $n \geq 1$, $q_B(C_4 \Box C_{2n+1}) \leq 5$.*

Proof. Set $q = 2n+1$ and use coordinates $(i,j) \in \mathbb{Z}_q \times \mathbb{Z}_4$, where i indexes the C_q factor. If $q \equiv 3 \pmod 4$, concatenate one copy of B_3 and $(q-3)/4$ copies of B_4. If $q \equiv 1 \pmod 4$, concatenate one copy of B_5 and $(q-5)/4$ copies of B_4. The final horizontal list of each block is 2315, and the initial vertical list of each block is 1234, so the same local transition occurs at every concatenation point.

Let $y_i(j)$ and $x_i(j)$ be the jth entries of Y_i and X_i, respectively. At vertex (i,j) the four incident colors are

$$\{y_i(j-1), y_i(j), x_{i-1}(j), x_i(j)\},$$

and the square with lower-left corner (i,j) has colors

$$\{x_i(j), x_i(j+1), y_i(j), y_{i+1}(j)\}.$$

Substitution in Table 1 shows that both sets have size four, including at every block boundary. Hence the concatenated coloring is a B-coloring. □

Corollary 3.4. *For all integers $n_1, n_2 \geq 1$, $q_B(C_{4n_1} \square C_{2n_2+1}) \leq 5$.*

Proof. Pull back the coloring in Lemma 3.3 through the covering $C_{4n_1} \to C_4$. □

For two odd factors, we use an explicit seam construction. The full seam data are given in Appendix A; only the defining formulas are needed in the proof below.

Proposition 3.5. *For all integers $n_1, n_2 \geq 1$,*

$$q_B(C_{2n_1+1} \square C_{2n_2+1}) \leq 5.$$

Proof. Set $p = 2n_1 + 1$ and $q = 2n_2 + 1$, and use coordinates $(i,j) \in \mathbb{Z}_p \times \mathbb{Z}_q$. Let $h(i,j)$ and $v(i,j)$ denote the colors of $(i,j)(i+1,j)$ and $(i,j)(i,j+1)$, respectively.

Suppose first that $q = 3$. For $0 \leq i \leq p-3$, define

$$\begin{aligned} h(i,0) &= (i \bmod 3) + 3, & v(i,0) &= ((i+1) \bmod 3) + 3,\\ h(i,1) &= 2 - (i \bmod 2), & v(i,1) &= ((i+2) \bmod 3) + 3,\\ h(i,2) &= ((i+1) \bmod 3) + 3, & v(i,2) &= 1 + (i \bmod 2). \end{aligned}$$

Let $H_i = (h(i,0), h(i,1), h(i,2))$ and $V_i = (v(i,0), v(i,1), v(i,2))$. The last two columns are specified in Table 2.

Table 2. The final two columns in the construction of $C_{2n_1+1} \square C_3$.

$p \bmod 6$	H_{p-2}	V_{p-2}	H_{p-1}	V_{p-1}
1	153	342	532	214
3	135	512	512	243
5	135	412	213	524

Checking the incident colors and the adjacent squares at the two seam columns shows that the coloring is proper and that every square is rainbow. Interchanging the two factors also covers $p = 3$.

For $p = q = 5$, identify the colors with $\mathbb{Z}_5$ and define $h(i,j) = i + j$ and $v(i,j) = i + j + 2$. At each vertex the four incident colors are $i+j-1, i+j, i+j+1, i+j+2$, and on each square they are $i+j, i+j+1, i+j+2, i+j+3$.

After interchanging the factors if necessary, the only remaining range is $p \geq 5$ and $q \geq 7$. Begin with

$$h_0(i,j) = 4 + ((i+j+1) \bmod 2), \qquad v_0(i,j) = ((j + 2 - (i \bmod 2)) \bmod 3) + 1.$$

Modify the terminal rows as listed in Appendix A. Then modify only columns $0, p-2, p-1$. For $\xi \in \{0, p-2, p-1\}$, write H_j^ξ and V_j^ξ for the replacement colors assigned to $h(\xi, j)$ and $v(\xi, j)$, respectively, and collect them as

$$T_j = (H_j^0, H_j^{p-2}, H_j^{p-1}; V_j^0, V_j^{p-2}, V_j^{p-1}),$$

the six entries prescribe, respectively, $h(0,j), h(p-2,j), h(p-1,j)$ and $v(0,j), v(p-2,j), v(p-1,j)$. The lists T_j are also given in Appendix A.

All remaining entries retain the values h_0, v_0. At every vertex it remains to verify

$$|\{h(i-1,j), h(i,j), v(i,j-1), v(i,j)\}| = 4, \tag{3.3}$$

and for every square

$$|\{h(i,j), h(i,j+1), v(i,j), v(i+1,j)\}| = 4. \tag{3.4}$$

Away from the terminal rows and columns, the periodic formulas give (3.3)–(3.4). At a seam, each check involves at most two consecutive rows and two consecutive columns. Substituting the finite lists in Appendix A verifies (3.3)–(3.4), including the two cyclic closures. Thus the displayed rules give a five-color B-coloring of $C_{2n_1+1}\square C_{2n_2+1}$. □

Proof of Theorem 3.1. Lemma 3.2 gives the lower bound five. For the matching upper bound, Proposition 3.5 applies when m is odd, and Corollary 3.4 applies when $m \equiv 0 \pmod 4$. In the remaining case, $m = 4a+2$ for some integer $a \geq 1$, and a five-coloring of $C_{2a+1}\square C_{2n+1}$ pulls back through the covering $C_{4a+2} \to C_{2a+1}$. □

Proof of Theorem 1.4. If at least one side length is odd, the result is Theorem 3.1. Suppose that m and n are even, with $m \leq n$. When $m = 4$ and $n = 4k+2$ for some integer $k \geq 1$, Theorem 1.2 gives $q_B(C_m \square C_n) = 5$. Every other even–even torus grid is non-exceptional, so the same theorem gives $q_B(C_m \square C_n) = 4$. □

Corollary 3.6. *Let $d \geq 1$ and $n_1, \ldots, n_d \geq 1$ be integers. Then*

$$q_B(C_{2n_1+1}\square \cdots \square C_{2n_d+1}) \leq \left\lceil \frac{5d}{2} \right\rceil.$$

Proof. Pair the first $2\lfloor d/2 \rfloor$ factors. By Theorem 3.1, each pair has a five-color B-coloring; any unpaired odd cycle has a three-color B-coloring. Theorem 2.1 gives chromatic number three for every nonempty product of these factors, so the hypotheses of Lemma 2.2 hold at each step. Repeated application of the lemma uses five colors per pair and three for the unpaired cycle, when present, for a total of $\lceil 5d/2 \rceil$ colors. □

4 Cylindrical grids

4.1 Odd circumference

Theorem 4.1 (Odd cylindrical grids)**.** *Let $n \geq 1$ and $m \geq 2$ be integers. Then*

$$q_B(C_{2n+1}\square P_m) = \begin{cases} 4, & 2 \leq m \leq n, \\ 5, & m \geq n+1. \end{cases}$$

Identify $V(C_{2n+1}\square P_m)$ with $\mathbb{Z}_{2n+1} \times \{0, 1, \dots, m-1\}$. In an edge-coloring, let $h_j(i)$ denote the color of $(i,j)(i+1,j)$ for $i \in \mathbb{Z}_{2n+1}$ and $0 \le j \le m-1$, and let $v_j(i)$ denote the color of $(i,j)(i,j+1)$ for $i \in \mathbb{Z}_{2n+1}$ and $0 \le j \le m-2$.

Lemma 4.2. *If m and n are integers satisfying $n \ge 2$ and $2 \le m \le n$, then $C_{2n+1}\square P_m$ has a four-color B-coloring.*

Proof. Identify the four colors with $\mathbb{Z}_2^2 = \{0, 1, 2, 3\}$, with addition interpreted bitwise. Define

$$v_0(i) = \begin{cases} 0, & 0 \le i \le 2n-2 \text{ and } i \text{ is even,} \\ 1, & 0 \le i \le 2n-2 \text{ and } i \text{ is odd,} \\ 2, & i = 2n-1, \\ 3, & i = 2n, \end{cases}$$

and

$$h_0(i) = \begin{cases} 3, & 0 \le i \le 2n-2 \text{ and } i \text{ is even,} \\ 2, & 0 \le i \le 2n-2 \text{ and } i \text{ is odd,} \\ 1, & i = 2n-1, \\ 2, & i = 2n. \end{cases}$$

Recursively set

$$h_{j+1}(i) = h_j(i) + v_j(i) + v_j(i+1), \tag{4.1}$$

and, whenever another vertical layer is needed,

$$v_{j+1}(i) = v_j(i) + h_{j+1}(i-1) + h_{j+1}(i). \tag{4.2}$$

Since the sum of three distinct elements of $\mathbb{Z}_2^2$ is the fourth element, (4.1) makes every new square rainbow and (4.2) makes every new interior vertex incident with four distinct colors.

The recurrence also gives the following explicit formulas. For $0 \le j \le n-1$,

$$v_j(i) = \begin{cases} 2 + ((i+j) \bmod 2), & 0 \le i < j, \\ (i-j) \bmod 2, & j \le i \le 2n-j-2, \\ 2 + ((i-2n+j+1) \bmod 2), & 2n-j-1 \le i \le 2n, \end{cases} \tag{4.3}$$

and, for $1 \le j \le n-1$,

$$h_j(i) = \begin{cases} (i+j+1) \bmod 2, & 0 \le i \le j-2, \\ 2, & i = j-1, \\ 3 - ((i-j) \bmod 2), & j \le i \le 2n-j-2, \\ 1 - ((i-2n+j+1) \bmod 2), & 2n-j-1 \le i \le 2n. \end{cases} \tag{4.4}$$

On each alternating interval, (4.1)–(4.2) preserve the displayed pattern. Substitution at the four breakpoints verifies the exceptional entries and completes the induction. Equations (4.3)–(4.4) also show that both boundary rows are properly colored. Restricting the construction to the first m rows proves the lemma. $\square$

Lemma 4.3. *If m and n are integers satisfying $n \ge 2$ and $m \ge n+1$, then $C_{2n+1}\square P_m$ has no four-color B-coloring.*

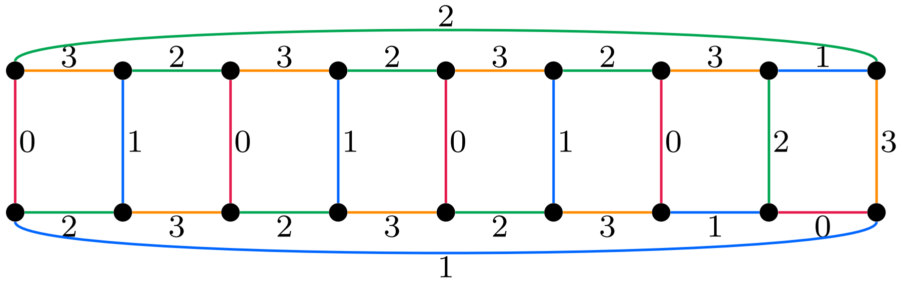

Figure 2. The initial two rows of the four-color construction for $C_9\Box P_m$, where m is an integer satisfying $2 \le m \le 4$. Edge colors are shown both by line color and by the labels $0, 1, 2, 3$.

Proof. Suppose that a four-color B-coloring exists. For a color c, let $A_j(c)$ be the number of horizontal edges of color c in row j, and let $B_j(c)$ be the number of vertical edges of color c between rows j and $j+1$. Fix c and abbreviate $A_j = A_j(c)$ and $B_j = B_j(c)$.

Counting color c on the squares between two consecutive rows gives, for $0 \le j \le m-2$,

$$A_j + A_{j+1} + 2B_j = 2n + 1, \tag{4.5}$$

while counting it on the edges incident with the vertices of an interior row gives, for $1 \le j \le m-2$,

$$2A_j + B_{j-1} + B_j = 2n + 1. \tag{4.6}$$

Comparing (4.5) for $j-1$ with (4.6) for j shows that $A_j + B_j$ is independent of j. Comparing (4.5) with (4.6) in the other direction shows that $A_{j+1} + B_j$ is also independent of j. The difference of these two constants is $A_{j+1} - A_j$, and hence $A_0, A_1, \dots, A_{m-1}$ is an arithmetic progression. Its common difference is odd, since the sum of the constants is $2n+1$.

The edges of color c in a cycle row form a matching, so $0 \le A_j \le n$. Since the progression has at least $n+1$ terms, its odd common difference must be 1 or -1, and the same bounds force $A_0(c) \in \{0, n\}$. Summing over the four colors in the first row gives a multiple of n, but that row has $2n+1$ edges, a contradiction. $\square$

Lemma 4.4. *For every integer $m \ge 2$, $C_3\Box P_m$ has no four-color B-coloring.*

Proof. It suffices to consider $C_3\Box P_2$. Relabel the colors so that one triangular layer has edge colors $1, 2, 3$ in cyclic order. Properness restricts the three joining edges to the sets $\{2,4\}$, $\{3,4\}$, and $\{1,4\}$, respectively. The square conditions leave only the joining triples

$$(2,3,1),\ (2,3,4),\ (2,4,1),\ (4,3,1).$$

The corresponding forced color triples on the second triangle are, respectively,

$$(4,4,4),\ (4,1,1),\ (3,3,4),\ (2,4,2),$$

none of which is proper. $\square$

Proof of Theorem 4.1. When $2 \le m \le n$, Lemma 4.2 gives the upper bound four, and the graph contains a 4-cycle. If $m \ge n+1$ and $n \ge 2$, Lemma 4.3 gives the lower bound five; the case $n = 1$ follows from Lemma 4.4. For the upper bound, embed P_m in any odd cycle of length at least $m+1$ and restrict the five-coloring supplied by Theorem 3.1. $\square$

4.2 Even circumference

Lemma 4.5. *For every integer $n \geq 2$, $q_B(P_2 \square C_{2n}) = 4$.*

Proof. Let $N = 2n$ and use coordinates $\{0,1\} \times \mathbb{Z}_N$. For $i \in \mathbb{Z}_N$, let d_i be the color of the P_2-edge in column i, and let x_i, y_i be the colors of the two cycle edges from column i to column $i+1$. Define

$$(d_i, x_i, y_i) = \begin{cases} (1,2,3), & i = 0, \\ (4,1,2), & i \text{ is odd and } 1 \leq i \leq N-3, \\ (3,2,1), & i \text{ is even and } 2 \leq i \leq N-2, \\ (4,3,2), & i = N-1. \end{cases} \tag{4.7}$$

For each i, the square between columns i and $i+1$ has colors x_i, y_i, d_i, d_{i+1}, namely all four colors, and the three colors incident with each vertex are distinct. This includes the cyclic closure from column $N-1$ to column 0, so (4.7) is a four-color B-coloring. The graph contains a 4-cycle, so four colors are necessary. □

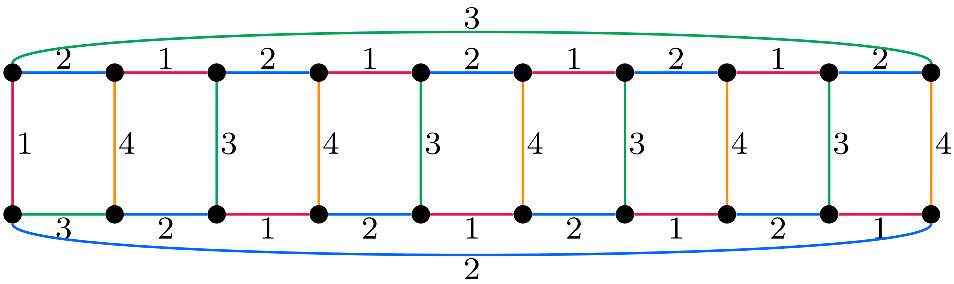

Figure 3. The four-color pattern (4.7) on $P_2 \square C_{10}$. The same two-column pattern is repeated for every even cycle length, with the two prescribed seam columns.

Proof of Theorem 1.5. The assertion for $C_{2n+1} \square P_m$ is Theorem 4.1. Now let $n \geq 2$. When $m = 2$, Lemma 4.5 gives $q_B(C_{2n} \square P_m) = 4$. Suppose that $m \geq 3$. If $n = 2$, then

$$C_4 \square P_m \cong P_2 \square P_2 \square P_m,$$

and Theorem 1.1 gives $q_B(C_4 \square P_m) = 4$. If $n \geq 3$, neither C_{2n} nor P_m contains a 4-cycle, and each has a proper two-edge-coloring; hence each has a two-color B-coloring. Both factors have chromatic number two, and Theorem 2.1 gives the same value for their product. Thus Lemma 2.2 applies and gives $q_B(C_{2n} \square P_m) \leq 4$. Since the product has maximum degree four, equality follows. □

5 Exceptional even discrete tori

The key step is a six-color construction that is uniform in two arbitrary odd cycle lengths.

Lemma 5.1. *For all integers $n_1, n_2 \geq 1$,*

$$q_B(P_2 \square P_2 \square C_{2n_1+1} \square C_{2n_2+1}) = 6.$$

Proof. Set $r = 2n_1 + 1$ and $q = 2n_2 + 1$. Let $G_{n_1,n_2} = P_2 \Box P_2 \Box C_{2n_1+1} \Box C_{2n_2+1}$ and identify

$$V(G_{n_1,n_2}) = \{0,1\}^2 \times \mathbb{Z}_r \times \mathbb{Z}_q.$$

Represent each vertex as $(\varepsilon, \delta, x, y)$, where $\varepsilon, \delta \in \{0,1\}$, $x \in \mathbb{Z}_r$, and $y \in \mathbb{Z}_q$. Denote the four edge directions by

$$\begin{aligned} e_0(\delta, x, y) &= (0, \delta, x, y)(1, \delta, x, y), \\ e_1(\varepsilon, x, y) &= (\varepsilon, 0, x, y)(\varepsilon, 1, x, y), \\ f(\varepsilon, \delta, x, y) &= (\varepsilon, \delta, x, y)(\varepsilon, \delta, x+1, y), \\ g(\varepsilon, \delta, x, y) &= (\varepsilon, \delta, x, y)(\varepsilon, \delta, x, y+1). \end{aligned}$$

We use the color set $\mathbb{Z}_6$. Addition of a scalar to a four-tuple is componentwise in $\mathbb{Z}_6$. For fixed (x, y), record the colors in the following order:

$$\begin{aligned} F_{x,y} &= (c(e_0(0, x, y)), c(e_1(1, x, y)), c(e_0(1, x, y)), c(e_1(0, x, y))), \\ X_{x,y} &= (c(f(0, 0, x, y)), c(f(1, 0, x, y)), c(f(0, 1, x, y)), c(f(1, 1, x, y))), \\ Y_{x,y} &= (c(g(0, 0, x, y)), c(g(1, 0, x, y)), c(g(0, 1, x, y)), c(g(1, 1, x, y))). \end{aligned}$$

The basic four-tuples are listed in Table 3. For $i \in \mathbb{Z}_3$, define $p_i = p + 2i$, $\bar{p}_i = \bar{p} + 2i$, $a_i = a + 2i$, $b_i = b + 2i$, $u_i = u + 2i$, and $v_i = v + 2i$.

Table 3. Basic four-tuples for the six-color construction; all entries are in $\mathbb{Z}_6$.

symbol	value	symbol	value
p	$(3, 4, 5, 2)$	$\bar{p}$	$(4, 5, 2, 3)$
d_0	$(2, 3, 4, 5)$	d_1	$(5, 2, 3, 4)$
d_2	$(3, 0, 1, 2)$	d_3	$(1, 4, 5, 0)$
a	$(0, 1, 3, 2)$	b	$(2, 3, 1, 0)$
u	$(1, 2, 0, 3)$	v	$(3, 0, 2, 1)$

Let $s = (r-3)/2$, so $s \geq 0$. For four-tuples U, V, let $(U, V)^s$ denote s consecutive copies of the pair U, V; this part is omitted when $s = 0$. We use four layer types, denoted by S_0, S_1, S_2, S_3. If the yth C_q-layer has type S_i, its F- and X-profiles are the two words in row S_i of Table 4, read in the order $x = 0, 1, \ldots, r-1$.

Table 4. The four layer types. Each profile has length $r = 2n_1 + 1 = 2s + 3$.

type	$(F_{x,y})_{x \in \mathbb{Z}_r}$	$(X_{x,y})_{x \in \mathbb{Z}_r}$
S_0	$(p_0, p_1, p_2, (p_0, d_0)^s)$	$(a_0, a_1, a_2, (a_0, a_2)^s)$
S_1	$(\bar{p}_0, \bar{p}_1, \bar{p}_2, (\bar{p}_0, d_1)^s)$	$(b_0, b_1, b_2, (b_0, b_2)^s)$
S_2	$(\bar{p}_2, \bar{p}_0, \bar{p}_1, (\bar{p}_2, d_2)^s)$	$(b_2, b_0, b_1, (b_2, b_1)^s)$
S_3	$(\bar{p}_1, \bar{p}_2, \bar{p}_0, (\bar{p}_1, d_3)^s)$	$(b_1, b_2, b_0, (b_1, b_0)^s)$

Table 5 gives the colors on the edges joining two consecutive layers. If a layer of type S_i is followed by one of type S_j, the corresponding row prescribes the Y-profile in the order $x = 0, 1, \ldots, r-1$.

Let $t = (q-3)/2$, so $t \geq 0$. Assign the following types to the C_q-layers in cyclic order:

$$S_1, \quad \underbrace{S_0, S_1, \ldots, S_0, S_1}_{t \text{ copies of the pair } S_0, S_1}, \quad S_2, \quad S_3.$$

Table 5. The Y-profiles on edges joining consecutive layers.

consecutive layer types	$(Y_{x,y})_{x\in\mathbb{Z}_r}$
$S_0 \to S_1,\ S_3 \to S_1$	$(u_0, u_1, u_2, (u_0, v_0)^s)$
$S_1 \to S_0,\ S_1 \to S_2$	$(u_2, u_0, u_1, (u_2, v_2)^s)$
$S_2 \to S_3$	$(u_1, u_2, u_0, (u_1, v_1)^s)$

The repeated part is omitted when $t = 0$. Every consecutive pair appears in Table 5, including S_3, S_1 at the cyclic closure, so the table completes the definition of the coloring. Figure 4 illustrates this layer order.

In each profile, the first three entries are prescribed separately and the rest alternate between two fixed four-tuples. It therefore suffices to check consecutive x-coordinates among the first three positions, at the junction with the alternating part, within its two possible adjacencies, and at the cyclic closure; these checks are independent of s.

For a vertex $(\varepsilon, \delta, x, y)$, substitution in Table 4 shows that the colors on

$$e_0(\delta, x, y), \quad e_1(\varepsilon, x, y), \quad f(\varepsilon, \delta, x-1, y), \quad f(\varepsilon, \delta, x, y)$$

are distinct. The same substitutions show that the three types of coordinate 4-cycle contained in one layer are rainbow. Next, for each row of Table 5, the color of a joining g-edge differs from the four non-g colors at both endpoints, and the three types of coordinate 4-cycle containing that edge are rainbow.

It remains to compare the two Y-profiles entering and leaving a layer. The cyclic order has at most seven local three-layer patterns: two in the repeated part and the others at its ends or around S_1, S_2, S_3. Entrywise comparison in Table 5 shows that the incoming and outgoing g-edges have different colors in every case.

Thus the six edges incident with every vertex have pairwise distinct colors. Since each factor is 4-cycle-free, every 4-cycle of G_{n_1,n_2} is a coordinate square and hence is rainbow by the preceding checks. This gives a six-color B-coloring, while $\Delta(G_{n_1,n_2}) = 6$ gives the reverse inequality. $\square$

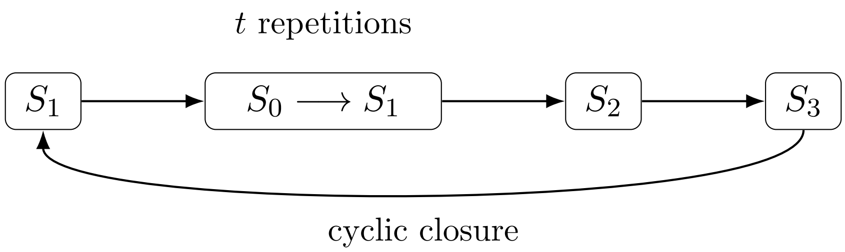

Figure 4. The cyclic order of the C_q-layers. Starting from S_1, the pair S_0, S_1 is repeated $t = (q-3)/2$ times, followed by S_2, S_3 and the return to S_1. The repeated part is absent when $t = 0$.

Corollary 5.2. *For all integers $n_1, n_2 \geq 1$,*

$$q_B(C_4 \square C_{4n_1+2} \square C_{4n_2+2}) = 6.$$

Proof. Use $P_2 \square P_2 \cong C_4$ and pull the coloring in Lemma 5.1 back through the double coverings $C_{4n_1+2} \to C_{2n_1+1}$ and $C_{4n_2+2} \to C_{2n_2+1}$. The lower bound is the maximum degree. $\square$

Proof of Theorem 1.6. The cases $d = 1$ and $d = 2$ are Theorem 1.2. Suppose $d \geq 3$. Since the discrete torus is exceptional, $C_{2n_1} = C_4$ and $n_2, \ldots, n_d$ are odd. By Corollary 5.2,

$$q_B(C_4 \square C_{2n_2} \square C_{2n_3}) = 6.$$

Every remaining even cycle has chromatic number and B-coloring number equal to two. By Theorem 2.1, the three-factor product above and every partial product obtained by adjoining further even cycles also have chromatic number two. Hence Lemma 2.2 applies repeatedly to the six-color base block and the two-color cycle factors, giving

$$q_B(C_{2n_1} \square \cdots \square C_{2n_d}) \leq 6 + 2(d-3) = 2d.$$

The product is $2d$-regular, so equality follows. □

6 Even ℓ-cylindrical grids

Together, Theorems 1.2 and 1.6 yield the following classification.

Corollary 6.1. *Let E be a Cartesian product of even cycles. Then $q_B(E) = \Delta(E)$ unless $E \cong C_4$ or $E \cong C_4 \square C_{4n+2}$ for some integer $n \geq 1$. In the two exceptional cases, $q_B(C_4) = 4$ and $q_B(C_4 \square C_{4n+2}) = 5$.*

Proof of Theorem 1.7. At a vertex of maximum degree, properness requires $\Delta(G)$ distinct colors; hence $q_B(G) \geq \Delta(G)$.

Item (1) is Lemma 4.5. Item (2) follows from Theorem 1.1, since $P_2 \square C_4 \square C_4 \cong Q_5$.

Every path and even-cycle factor has chromatic number two, and Theorem 2.1 gives the same value for every nontrivial product formed from them. Since all B-colorings used below have at least two colors, Lemma 2.2 applies whenever invoked.

We next prove the upper bounds in item (3). In (3a), select one cycle factor of E. Lemma 4.5 colors its product with P_2 using four colors, while the remaining even-cycle product has B-coloring number equal to its maximum degree because it contains no C_4. Lemma 2.2 now gives $q_B(G) \leq \Delta(G)+1$. In (3b), apply the same lemma to $P_2 \square C_4 \square C_4 \cong Q_5$ and C_{4n+2} to obtain the upper bound $6 + 2 = \Delta(G) + 1$. In (3c), Theorem 1.1 gives $q_B(P_2 \square P_m) = 4 = \Delta(P_2 \square P_m) + 1$, while $q_B(E) = \Delta(E)$ because E has no C_4. Another application of Lemma 2.2 gives $\Delta(G) + 1$.

It remains to prove item (4). Let $k = c_4(E)$.

First suppose that P is neither P_2 nor $P_2 \square P_m$. If E is not one of the two exceptions in Corollary 6.1, apply Lemma 2.2 directly to P and E. If $E \cong C_4$, regard $P \square C_4$ as a product of paths by replacing C_4 with $P_2 \square P_2$; the resulting path product is not one of the exceptions in Theorem 1.1. If $E \cong C_4 \square C_{4n+2}$, first color $P \square C_4$ in this way and then add the remaining cycle using Lemma 2.2. In each case the number of colors is $\Delta(G)$.

Suppose next that $P \cong P_2 \square P_m$ and $k \geq 1$. Absorb all k copies of $C_4 \cong P_2 \square P_2$ into the path part. The resulting product $P_2^{2k+1} \square P_m$ is not a path-product exception, while the remaining even-cycle product contains no C_4. Theorem 1.1 gives the required coloring if no even-cycle factor remains. Otherwise, Theorems 1.1 and 1.2, followed by Lemma 2.2, give $q_B(G) = \Delta(G)$.

It remains to consider $P \cong P_2$. The cases $t = 1$, $E \cong C_4^2$, $k = 0$ with $t \geq 2$, and $E \cong C_4^2 \square C_{4n+2}$ have already been listed. In the next two cases, H is connected, has at least two edges, and Theorem 2.1 gives $\chi(H) = 2$; hence it is bipartite, as required in Lemma 2.3.

If $k = 1$, let H be the nonempty product obtained by deleting the unique C_4 factor, so that $G = Q_3 \square H$. Since H contains no C_4, Corollary 6.1 and Lemma 2.3 give $q_B(G) = \Delta(G)$. If $k = 2$ and the graph is not one of the listed cases, retain one C_4 factor in H and again express G as $Q_3 \square H$. The product H is neither C_4 nor $C_4 \square C_{4n+2}$, so Corollary 6.1 and Lemma 2.3 apply. If $k \geq 3$, absorb all C_4 factors into the path part, obtaining the hypercube Q_{2k+1}. Since $2k + 1 \geq 7$, Theorem 1.1 gives $q_B(Q_{2k+1}) = 2k + 1$. This is the desired coloring when no even-cycle factor remains. Otherwise, the remaining even-cycle product contains no C_4, and Lemma 2.2 completes the proof. $\square$

7 Open problems

Theorems 1.4 and 1.6 exhibit two different parity phenomena. A single odd factor forces five colors in a two-dimensional cycle product, whereas the one-color obstruction for exceptional even discrete tori disappears as soon as the dimension is at least three. Corollary 3.6 also gives, for every positive integer d and every product G of d odd cycles,

$$2d + 1 \leq q_B(G) \leq \left\lceil \frac{5d}{2} \right\rceil .$$

The lower bound is attained under the divisibility hypothesis in Theorem 1.3. This suggests the following conjecture.

Conjecture 7.1. *For all integers $d \geq 1$ and $n_1, \ldots, n_d \geq 1$,*

$$q_B(C_{2n_1+1} \square \cdots \square C_{2n_d+1}) = 2d + 1.$$

The three interval classes in Theorem 1.7(3) are the remaining unresolved even ℓ-cylindrical grids. Determining which of these grids have B-coloring number Δ and which require $\Delta + 1$ is the second open problem arising from our classification.

A Seam data for the odd–odd construction

This appendix gives the finite lists used in Proposition 3.5. For the integers $n_1, n_2 \geq 1$ in that proposition, set $p = 2n_1 + 1$ and $q = 2n_2 + 1$. Write $h^\circ(i,j)$ and $v^\circ(i,j)$ for the replacement values assigned to $h_0(i,j)$ and $v_0(i,j)$ in the terminal rows. Each ordered pair in Table 6 is $(h^\circ(i,j), v^\circ(i,j))$, and the row is chosen according to the parity of i.

Table 6. Corrections in the terminal rows of the periodic coloring.

$q \pmod 6$	$i \pmod 2$	terminal rows, from left to right		
1	0	$(2,3)$ at $q-2$	$(4,1)$ at $q-1$	
1	1	$(5,1)$ at $q-2$	$(2,3)$ at $q-1$	
3	0	$(3,1)$ at $q-3$	$(2,5)$ at $q-2$	$(4,1)$ at $q-1$
3	1	$(4,5)$ at $q-3$	$(3,1)$ at $q-2$	$(2,3)$ at $q-1$
5	0	$(3,1)$ at $q-2$	$(4,2)$ at $q-1$	
5	1	$(5,2)$ at $q-2$	$(3,1)$ at $q-1$	

For the three terminal columns, recall that

$$T_j = (H_j^0, H_j^{p-2}, H_j^{p-1}; V_j^0, V_j^{p-2}, V_j^{p-1}).$$

The two short cases are listed in Table 7.

Table 7. The terminal-column lists for $q = 7$ and $q = 9$.

j	T_j for $q=7$	T_j for $q=9$
0	$(1,1,2;3,2,4)$	$(1,1,2;3,2,4)$
1	$(4,5,1;5,3,2)$	$(4,5,1;5,3,2)$
2	$(2,4,3;4,1,5)$	$(2,4,3;4,1,5)$
3	$(3,3,1;5,2,4)$	$(3,3,1;5,2,4)$
4	$(1,1,3;4,4,2)$	$(1,1,2;4,3,5)$
5	$(2,3,5;3,1,4)$	$(2,2,1;5,1,3)$
6	$(5,2,1;4,3,5)$	$(3,4,2;4,5,1)$
7	—	$(2,3,5;3,1,4)$
8	—	$(5,2,1;4,3,5)$

For $q \geq 11$, let

$$E_j = \begin{cases} 5, & j \text{ even}, \\ 4, & j \text{ odd}, \end{cases} \qquad A = (3,1,2), \qquad B = (2,3,1), \qquad C = (1,2,3),$$

where the entries of A, B, and C are indexed by $\mathbb{Z}_3$. In every indicated middle range, set

$$T_j = (E_j, A_{j \bmod 3}, B_{j \bmod 3}; C_{j \bmod 3}, B_{j \bmod 3}, E_j). \tag{A.1}$$

The initial lists are given in Table 8; the terminal lists are given in Table 9.

Table 8. Initial terminal-column lists for $q \geq 11$.

j	$q \equiv 1,3 \pmod 6$	$q \equiv 5 \pmod 6$
0	$(1,1,2;3,2,4)$	$(3,1,2;1,2,4)$
1	$(4,5,1;5,3,2)$	$(4,5,3;2,3,1)$
2	$(2,4,3;4,1,5)$	$(5,2,4;3,1,5)$
3	$(3,3,1;5,2,4)$	$(4,3,1;5,2,4)$
4	$(1,1,2;4,3,5)$	$(1,1,2;4,3,5)$
5	$(2,2,1;3,1,4)$	$(2,2,1;3,1,4)$

Table 9. Middle ranges and terminal-column lists for $q \geq 11$.

$q \pmod 6$	middle range for (A.1)	index	T_j
1	$6 \leq j \leq q-4$	$q-3$	$(5,1,3;4,4,2)$
		$q-2$	$(2,3,5;3,1,4)$
		$q-1$	$(5,2,1;4,3,5)$
3	$6 \leq j \leq q-5$	$q-4$	$(4,2,1;3,5,4)$
		$q-3$	$(2,1,5;1,4,2)$
		$q-2$	$(4,5,3;2,1,4)$
		$q-1$	$(5,2,1;4,3,5)$
5	$6 \leq j \leq q-4$	$q-3$	$(5,2,1;3,4,5)$
		$q-2$	$(4,1,2;1,5,4)$
		$q-1$	$(5,2,3;4,3,5)$

References

[1] Anita Arora. "The monopole-dimer model on high-dimensional cylindrical, toroidal, Möbius and Klein grids". In: *Séminaire Lotharingien de Combinatoire* 93B (2025), Article 58, 12 pp. URL: https://www.mat.univie.ac.at/~slc/wpapers/FPSAC2025/58.pdf.

[2] W. G. Brown, P. Erdős, and V. T. Sós. "Some extremal problems on r-graphs". In: *New Directions in the Theory of Graphs*. New York: Academic Press, 1973, pp. 55–63.

[3] Dong Chen, Weifan Wang, and Yitong Xu. "B-coloring of $\mathcal{MP}_2$-graphs". In: *Applied Mathematics and Computation* 531 (2026), p. 130186. DOI: 10.1016/j.amc.2026.130186.

[4] Zdeněk Dvořák, Bojan Mohar, and Robert Šámal. "Star chromatic index". In: *Journal of Graph Theory* 72.3 (2013), pp. 313–326. DOI: 10.1002/jgt.21644.

[5] Pál Erdős and András Gyárfás. "A variant of the classical Ramsey problem". In: *Combinatorica* 17.4 (1997), pp. 459–467. DOI: 10.1007/BF01195056.

[6] Ralph J. Faudree, András Gyárfás, Linda Lesniak, and Richard H. Schelp. "Rainbow coloring of the cube". In: *Journal of Graph Theory* 17.5 (1993), pp. 607–612. DOI: 10.1002/jgt.3190170510.

[7] Ralph J. Faudree, András Gyárfás, and Richard H. Schelp. "An edge coloring problem for graph products". In: *Journal of Graph Theory* 23.3 (1996), pp. 297–302. DOI: 10.1002/(SICI)1097-0118(199611)23:3<297::AID-JGT9>3.0.CO;2-D.

[8] András Gyárfás, Ryan R. Martin, Miklós Ruszinkó, and Gábor N. Sárközy. "Proper edge colorings of planar graphs with rainbow C_4's". In: *Journal of Graph Theory* 107.4 (2024), pp. 833–846. DOI: 10.1002/jgt.23163.

[9] András Gyárfás and Gábor N. Sárközy. ""Less" strong chromatic indices and the (7, 4)-conjecture". In: *Studia Scientiarum Mathematicarum Hungarica* 60.2–3 (2023), pp. 109–122. DOI: 10.1556/012.2023.01539.

[10] András Gyárfás and Gábor N. Sárközy. "Proper edge colorings of Cartesian products with rainbow C_4's". In: *Graphs and Combinatorics* 39.5 (2023), p. 98. DOI: 10.1007/s00373-023-02697-x.

[11] András Gyárfás, Gábor N. Sárközy, and Adam Zsolt Wagner. "Perfect proper edge colorings of regular bipartite graphs with rainbow C_4's". In: *Discrete Mathematics* 349.6 (2026), p. 115005. DOI: 10.1016/j.disc.2026.115005.

[12] Jiangxu Kong, Yiqiao Wang, and Mengmeng Zheng. "B-coloring of planar graphs". In: *Journal of Graph Theory* (2026). DOI: 10.1002/jgt.70067.

[13] Gert Sabidussi. "Graphs with given group and given graph-theoretical properties". In: *Canadian Journal of Mathematics* 9 (1957), pp. 515–525. DOI: 10.4153/CJM-1957-060-7.

[14] Yu Xue, Xiaoxue Hu, and Jiangxu Kong. "Proper edge coloring of subcubic graphs with rainbow C_4's". SSRN preprint. 2026. DOI: 10.2139/ssrn.6528922.